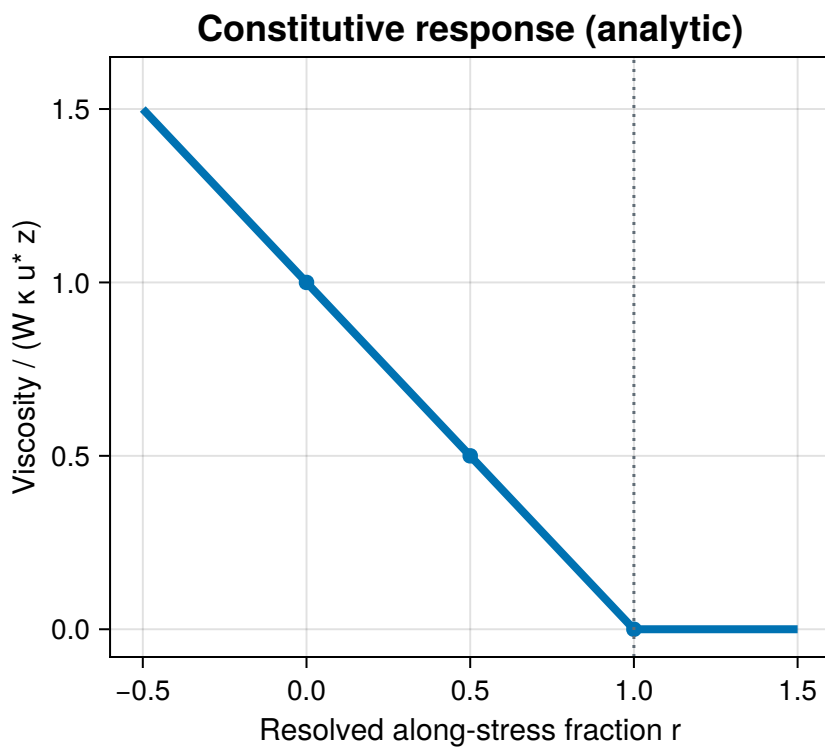


Can a thin surface closure improve coarse LES?

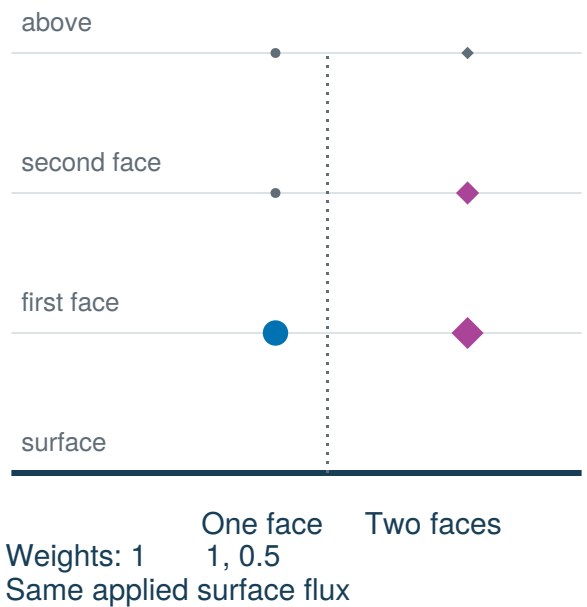
Supply unresolved near-wall transport while preserving resolved turbulence.
Status: CPU implementation evidence available; SLD LES comparisons pending.

A local estimate of missing transport

Filter velocity, scalar, and wall-flux samples in physical time. Use their centered covariance to estimate the resolved share of the surface flux.



Shallow vertical support



What this formula does

No resolved stress: full neutral similarity viscosity. Half resolved: half the viscosity. Fully resolved or more: the correction turns off.
Opposite-sign transport can give a factor greater than one.

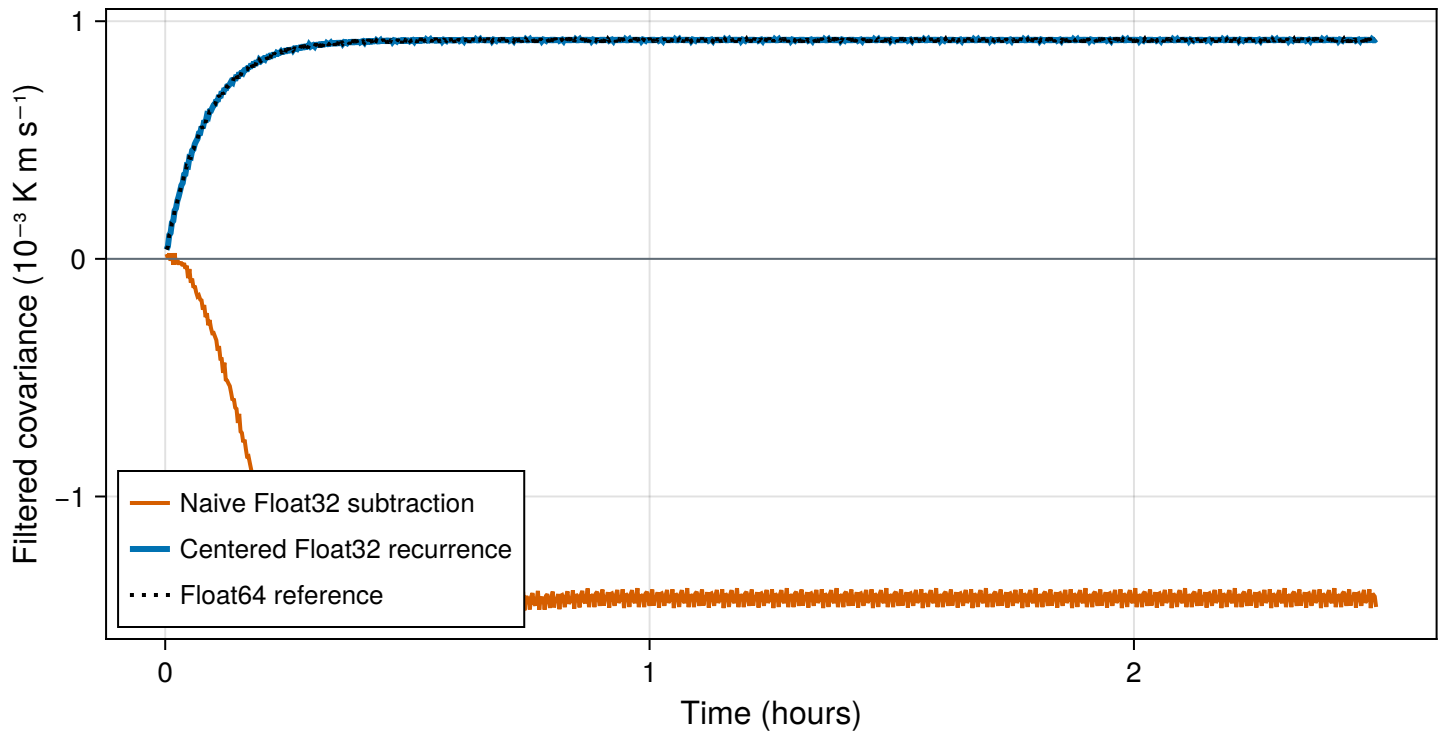
What it does not establish

The added flux still depends on the instantaneous vertical gradient.
The formula does not force exact flux balance, include stability functions, or measure the numerical transport produced by WENO.

One-face height: 12.5 m in both planned GABLS comparisons.
Scalar correction uses signed flux ratios; zero/tiny wall flux disables it.

A small covariance can get the wrong sign in Float32

Synthetic arithmetic test: 90,000 samples, $\Delta t = 0.1$ s, filter time = 300 s.
Identical Float32 input samples are used in the Float64 reference.



Final covariance: naive -1.465 | centered 0.925 | reference 0.924

Units: $10^{-3} \text{ K m s}^{-1}$. Centered recurrence relative error: 0.020%.

Why this matters for the closure

Subtracting two much larger filtered products loses the small turbulent flux. Here it reverses its sign. A centered covariance update retains the signal that controls the added diffusivity.

Implementation checks support the arithmetic result

119 CPU feature checks cover filtering, flux signs and guards, support, implicit conservation, and checkpoint continuation. Separate CPU runner and export checks cover shapes, native heights, schedules and provenance.

Boundary of the evidence: synthetic arithmetic and CPU checks are not evidence of improved LES fidelity. GPU validation and matched LES remain pending. Reproducer, exact plotted CSV, and source hashes accompany this section.

Judge transport, turbulence, and the mean state together

One controlled intervention, three complementary tests of physical behavior.
No completed SurfaceLayerDiffusivity LES result is being claimed here.

01 Did the near-wall transport change as intended?

Lead with resolved, added, and combined momentum/heat flux profiles.
Zoom the lowest four levels; mark every active face. Pair flux mismatch with viscosity, diffusivity, guard activity, and mean vertical transport.

02 Did the boundary layer improve without losing turbulence?

Place mean wind and temperature beside w^2 , w^3 and skewness. Show boundary-layer depth and surface flux through time. Retain the fixed 1 m GABLS1 median on comparable panels; gray bands show archive spread.

03 Is the result robust to filter time and support?

Use the same four styles in every GABLS panel. Compare with a freshly run, same-revision control, then examine 7-8 h versus 8-9 h. Report changes in physical units; avoid a composite score that hides tradeoffs.

-  Matched WENO9 control
-  One face / 100 s
-  One face / 300 s
-  Two faces / 300 s

Independent challenges strengthen the conclusion

GABLS3: moist nocturnal evolution and the sunrise flux sign change.
Neutral ABL: fixed-stress transport, without a surface-drag prediction.
Both add evidence; neither substitutes for the GABLS1 reference comparison.